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I. An Ancient Ordering

time limit per test: 1 second

memory limit per test: 256 megabytes

You have found an old dictionary containing N words. The words are stored in an order that is different from the regular Latin lexicographic order.

Your task is to determine the order of the alphabet that satisfies the lexicographic order of this dictionary. If there are multiple valid orders, print the **lexicographically smallest** one. For example, the sequence $S_1 = "d x i k"$ is lexicographically smaller than the sequence $S_2 = "d x p a k"$.

If no such valid sequence exists, print -1 . A valid ordering is not possible if the characters create cyclic dependencies or if a longer word appears before a shorter word that is a prefix of it.

Input

The first line contains an integer N ($1 \leq N \leq 1000$) — the number of words in the dictionary.

The next N line contains a string S ($1 \leq |S| \leq 100$). Each word consists of only lowercase Latin letters $a - z$.

Output

Find out the order of the alphabets that satisfy the sorting order of the words in the given dictionary. If there are multiple valid orders, print the **lexicographically smallest** one. If no such valid sequence exists, print -1 .

Examples

input
[Copy](#)

```
3
eat
tea
ate
```

output
[Copy](#)

```
eta
```

input
[Copy](#)

```
9
error
tooth
tot
teeth
their
there
thi
tie
hit
```

output
[Copy](#)

```
oethir
```

CSE221 Assignment 06 Summer 2026

Contest is running

3 days

Contestant


[→ Submit?](#)

 Language: Java 21 64bit

Choose file:

[Choose File](#)

No file chosen


[Submit](#)

	input	Copy
6 gef gie hf hd hc ha		
	output	Copy
efdcaghi		

input	Copy
5 cmwaqe yent jtdgx wlp xufjpf	
output	Copy
acdefglmpqtuyjwx	

input	Copy
6 abc ab p pq pqr pqrs	
output	Copy
-1	

input	Copy
2 pigeon pigeons	
output	Copy
eginops	

input	Copy
4 ab bc ca ac	
output	Copy
-1	

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The only programming contests Web 2.0 platform
Server time: Aug/20/2026 17:19:13^{UTC+6} (i1).
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